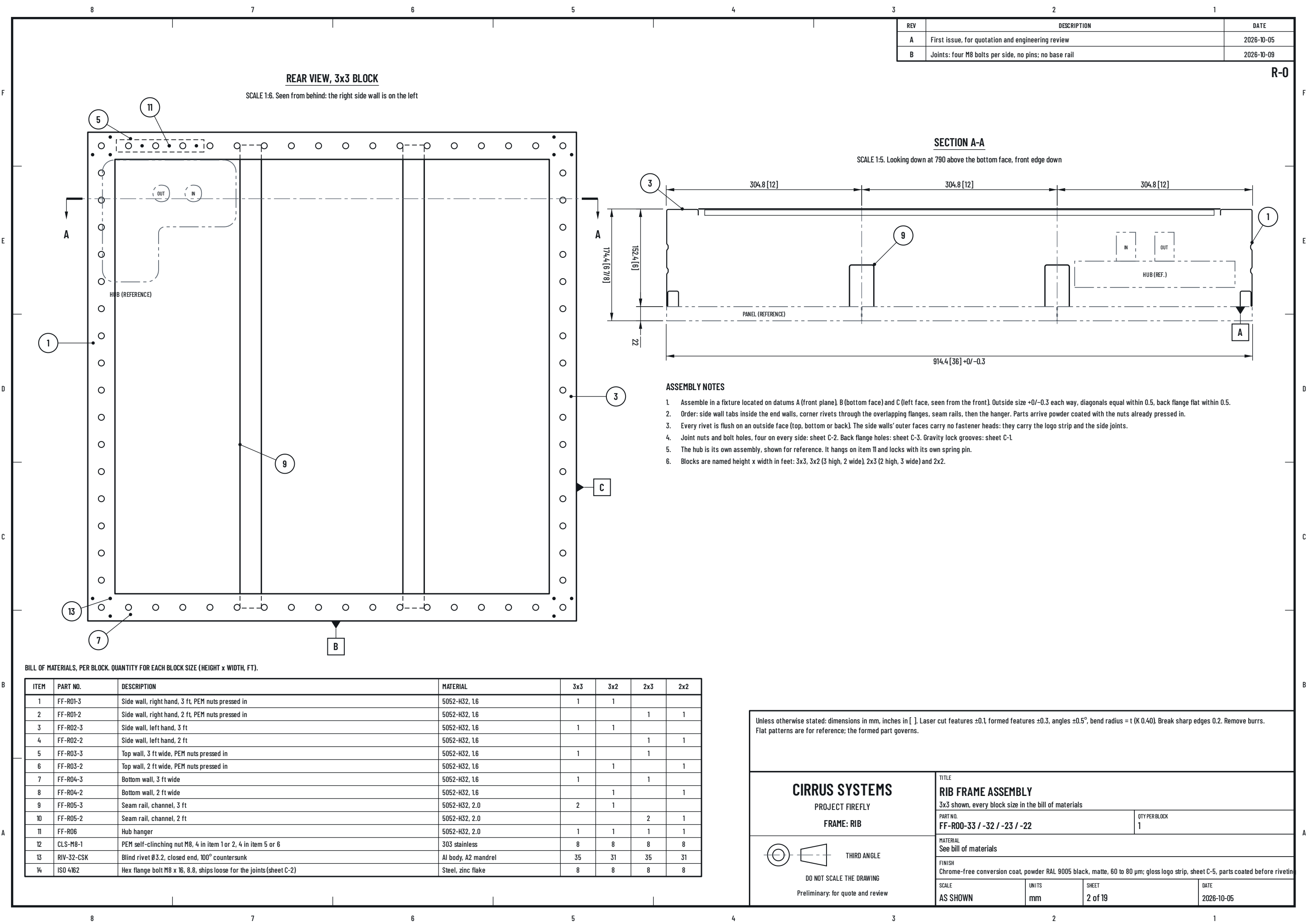
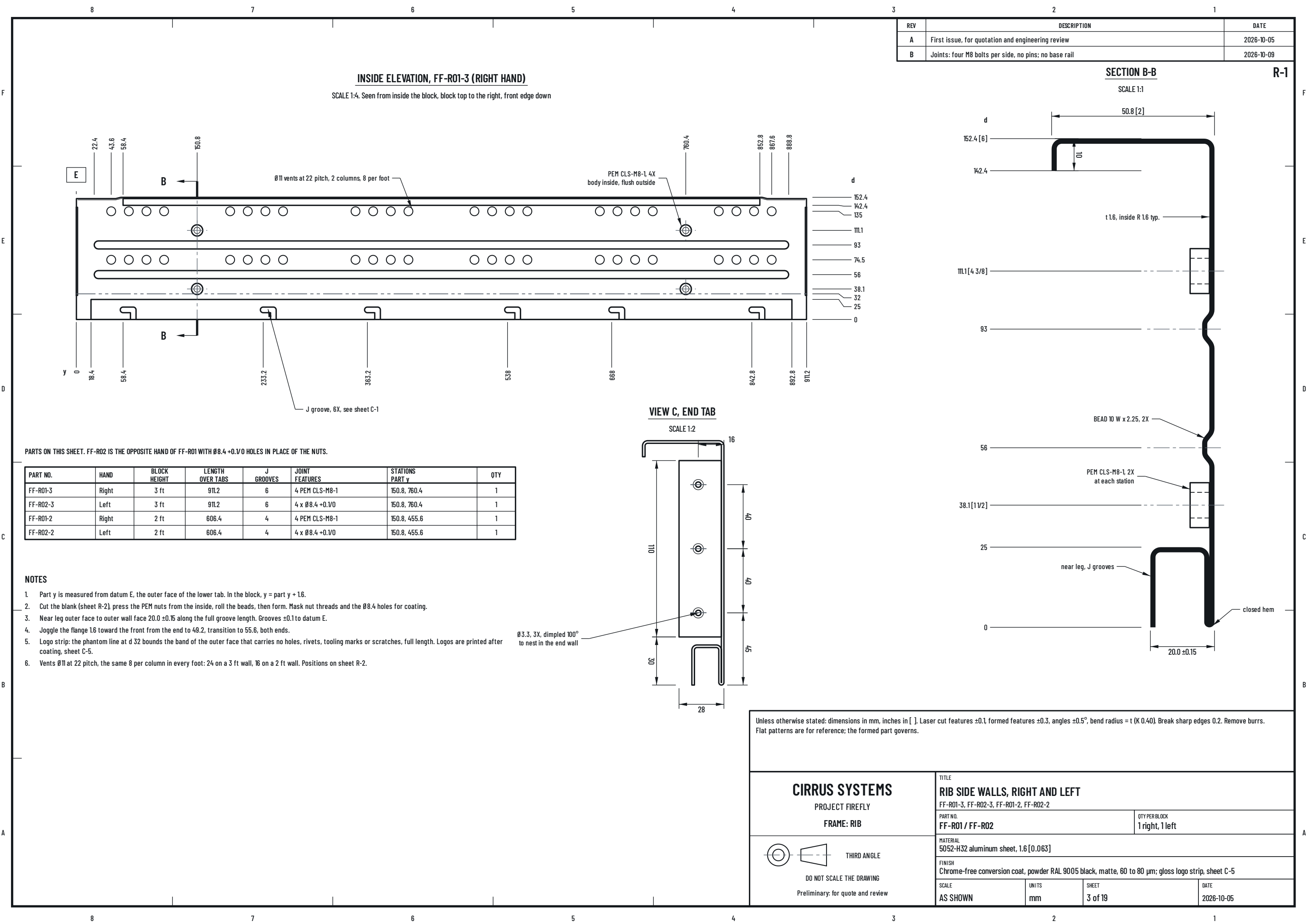
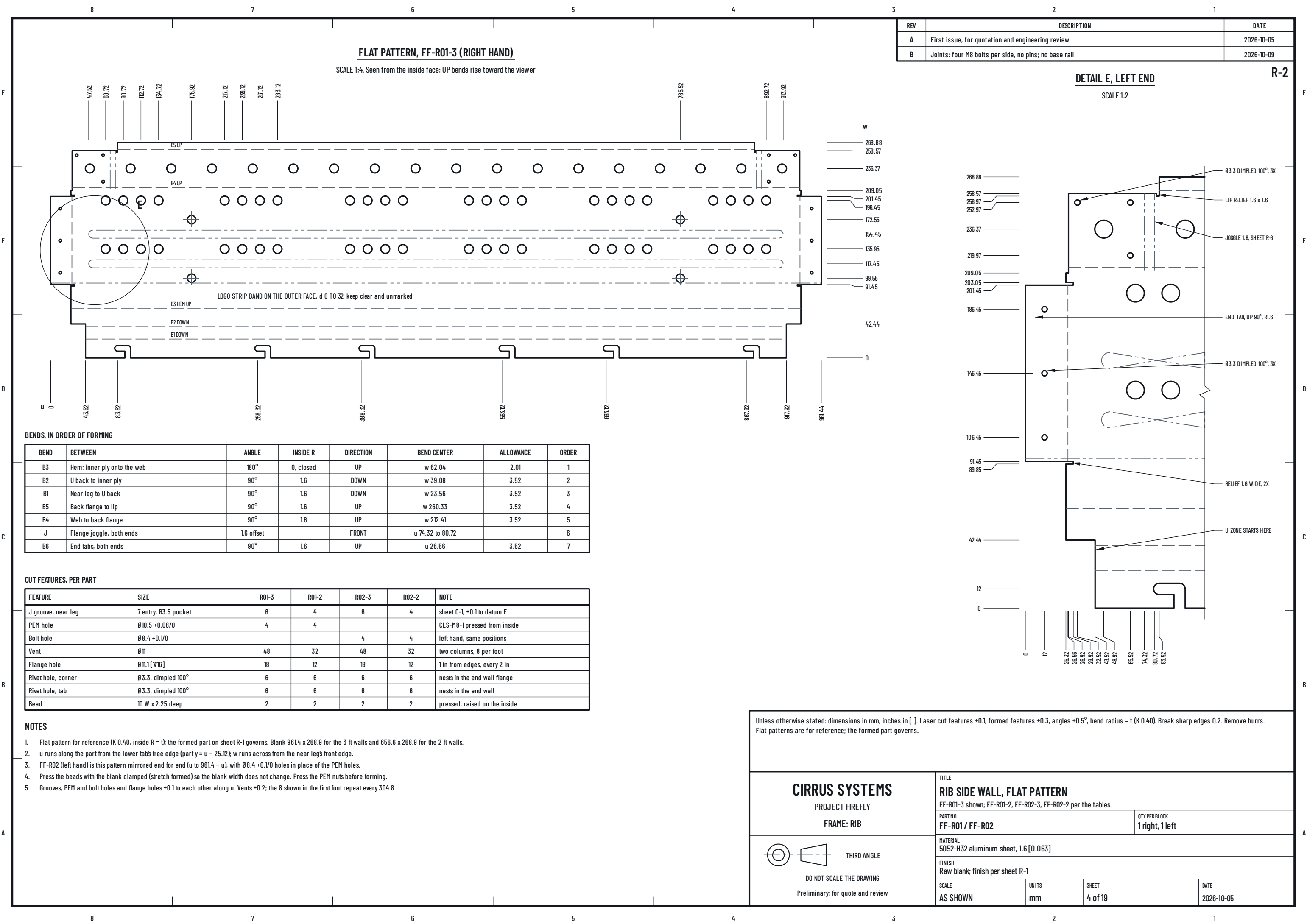


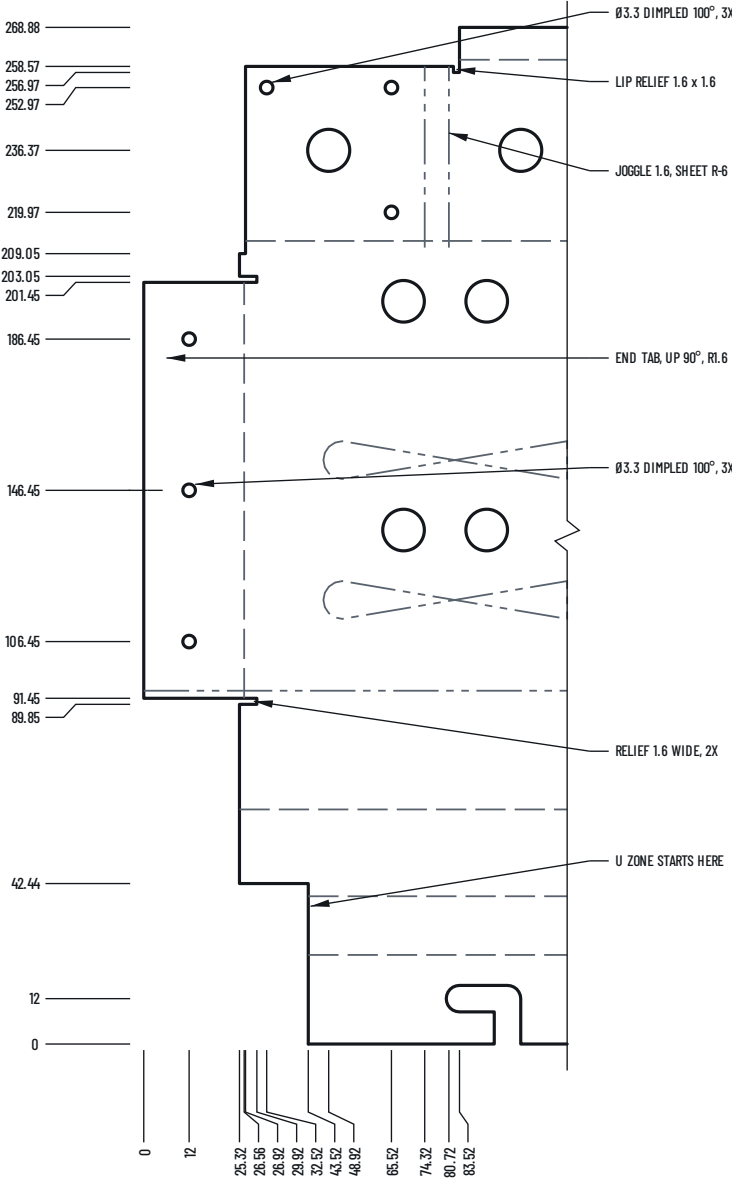






REV	DESCRIPTION	DATE
A	First issue, for quotation and engineering review	2026-10-05
B	Joints: four M8 bolts per side, no pins; no base rail	2026-10-09

DETAIL E, LEFT END
SCALE 1:2



BENDS, IN ORDER OF FORMING

BEND	BETWEEN	ANGLE	INSIDE R	DIRECTION	BEND CENTER	ALLOWANCE	ORDER
B3	Hem: inner ply onto the web	180°	0, closed	UP	w 62.04	2.01	1
B2	U back to inner ply	90°	1.6	DOWN	w 39.08	3.52	2
B1	Near leg to U back	90°	1.6	DOWN	w 23.56	3.52	3
B5	Back flange to lip	90°	1.6	UP	w 260.33	3.52	4
B4	Web to back flange	90°	1.6	UP	w 212.41	3.52	5
J	Flange joggle, both ends	1.6 offset		FRONT	u 74.32 to 80.72		6
B6	End tabs, both ends	90°	1.6	UP	u 26.56	3.52	7

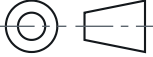
CUT FEATURES, PER PART

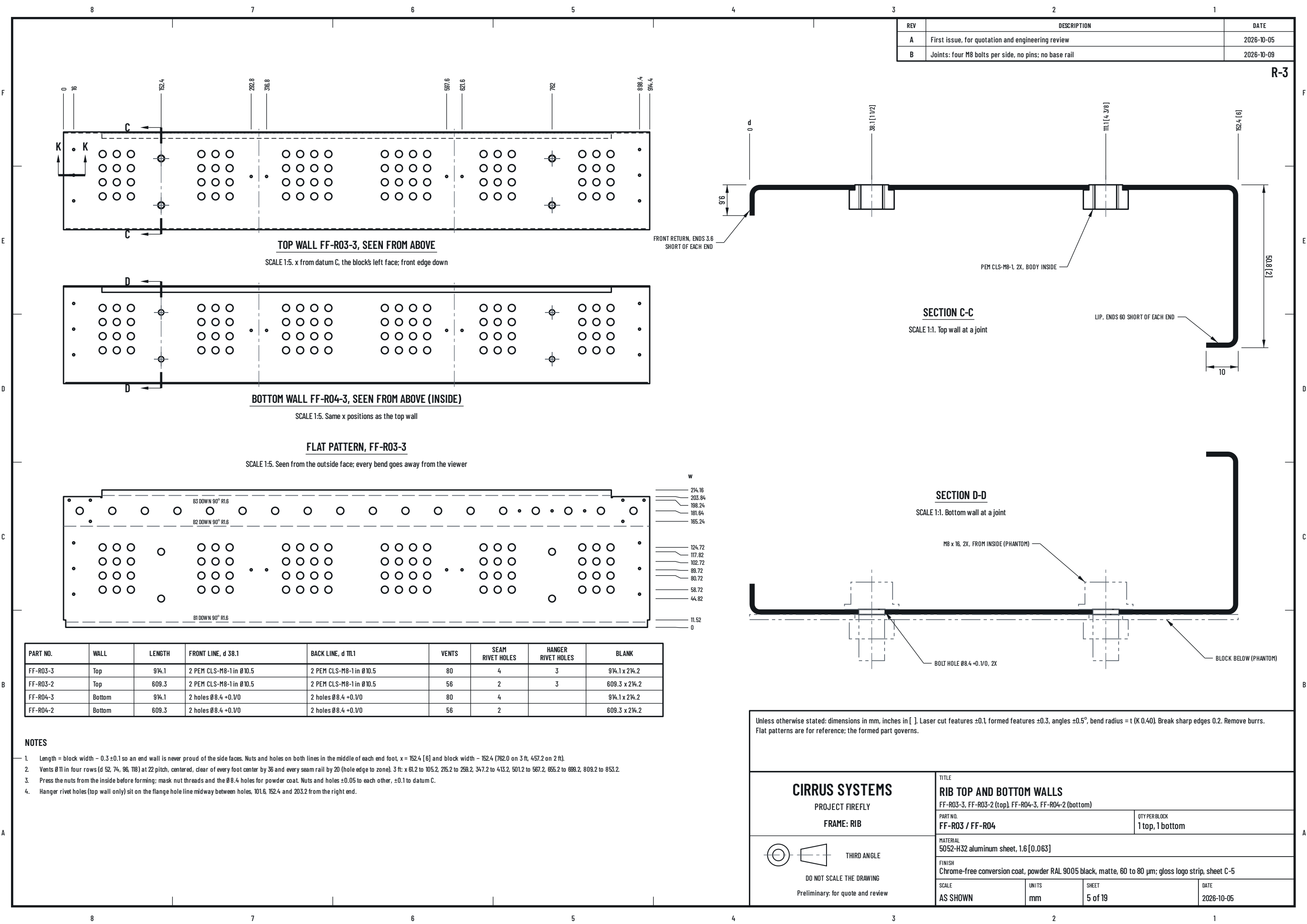
FEATURE	SIZE	R01-3	R01-2	R02-3	R02-2	NOTE
J groove, near leg	7 entry, R3.5 pocket	6	4	6	4	sheet C-1, ±0.1 to datum E
PEM hole	Ø10.5 +0.08/0	4	4			CLS-M8-1 pressed from inside
Bolt hole	Ø8.4 +0.1/0			4	4	left hand, same positions
Vent	Ø11	48	32	48	32	two columns, 8 per foot
Flange hole	Ø11.1 [7/16]	18	12	18	12	1 in from edges, every 2 in
Rivet hole, corner	Ø3.3, dimpled 100°	6	6	6	6	nests in the end wall flange
Rivet hole, tab	Ø3.3, dimpled 100°	6	6	6	6	nests in the end wall
Bead	10 W x 2.25 deep	2	2	2	2	pressed, raised on the inside

NOTES

- Flat pattern for reference (K 0.40, inside R = t): the formed part on sheet R-1 governs. Blank 961.4 x 268.9 for the 3 ft walls and 656.6 x 268.9 for the 2 ft walls.
- u runs along the part from the lower tabs free edge (part y = u - 25.12); w runs across from the near legs front edge.
- FF-R02 (left hand) is this pattern mirrored end for end (u to 961.4 - u), with Ø8.4 +0.1/0 holes in place of the PEM holes.
- Press the beads with the blank clamped (stretch formed) so the blank width does not change. Press the PEM nuts before forming.
- Grooves, PEM and bolt holes and flange holes ±0.1 to each other along u. Vents ±0.2; the 8 shown in the first foot repeat every 304.8.

Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ±0.1, formed features ±0.3, angles ±0.5°, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.

CIRRUS SYSTEMS PROJECT FIREFLY FRAME: RIB THIRD ANGLE DO NOT SCALE THE DRAWING Preliminary: for quote and review	TITLE RIB SIDE WALL, FLAT PATTERN FF-R01-3 shown; FF-R01-2, FF-R02-3, FF-R02-2 per the tables	
	PART NO. FF-R01 / FF-R02	QTY PER BLOCK 1 right, 1 left
	MATERIAL 5052-H32 aluminum sheet, 1.6 [0.063]	
	FINISH Raw blank; finish per sheet R-1	
	SCALE AS SHOWN	UNITS mm
SHEET 4 of 19		DATE 2026-10-05



REV	DESCRIPTION	DATE
A	First issue, for quotation and engineering review	2026-10-05
B	Joints: four M8 bolts per side, no pins; no base rail	2026-10-09

R-3

SECTION C-C

SCALE 1:1. Top wall at a joint

SECTION D-D

SCALE 1:1. Bottom wall at a joint

PART NO.	WALL	LENGTH	FRONT LINE, d 38.1	BACK LINE, d 111.1	VENTS	SEAM RIVET HOLES	HANGER RIVET HOLES	BLANK
FF-R03-3	Top	914.1	2 PEM CLS-M8-1 in Ø10.5	2 PEM CLS-M8-1 in Ø10.5	80	4	3	914.1 x 214.2
FF-R03-2	Top	609.3	2 PEM CLS-M8-1 in Ø10.5	2 PEM CLS-M8-1 in Ø10.5	56	2	3	609.3 x 214.2
FF-R04-3	Bottom	914.1	2 holes Ø8.4 +0.1/0	2 holes Ø8.4 +0.1/0	80	4		914.1 x 214.2
FF-R04-2	Bottom	609.3	2 holes Ø8.4 +0.1/0	2 holes Ø8.4 +0.1/0	56	2		609.3 x 214.2

NOTES

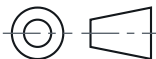
- Length = block width - 0.3±0.1 so an end wall is never proud of the side faces. Nuts and holes on both lines in the middle of each end foot, x = 152.4 [6] and block width - 152.4 (762.0 on 3 ft, 457.2 on 2 ft).
- Vents Ø11 in four rows (d 52, 74, 96, 118) at 22 pitch, centered, clear of every foot center by 36 and every seam rail by 20 (hole edge to zone). 3 ft: x 61.2 to 105.2, 215.2 to 258.2, 347.2 to 413.2, 501.2 to 567.2, 655.2 to 698.2, 809.2 to 853.2.
- Press the nuts from the inside before forming; mask nut threads and the Ø8.4 holes for powder coat. Nuts and holes ±0.05 to each other, ±0.1 to datum C.
- Hanger rivet holes (top wall only) sit on the flange hole line midway between holes, 101.6, 152.4 and 203.2 from the right end.

Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ±0.1, formed features ±0.3, angles ±0.5°, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.

CIRRUS SYSTEMS

PROJECT FIREFLY

FRAME: RIB



THIRD ANGLE

DO NOT SCALE THE DRAWING

Preliminary: for quote and review

TITLE

RIB TOP AND BOTTOM WALLS

FF-R03-3, FF-R03-2 (top), FF-R04-3, FF-R04-2 (bottom)

PART NO.

FF-R03 / FF-R04

QTY PER BLOCK

1 top, 1 bottom

MATERIAL

5052-H32 aluminum sheet, 1.6 [0.063]

FINISH

Chrome-free conversion coat, powder RAL 9005 black, matte, 60 to 80 µm; gloss logo strip, sheet C-5

SCALE

AS SHOWN

UNITS

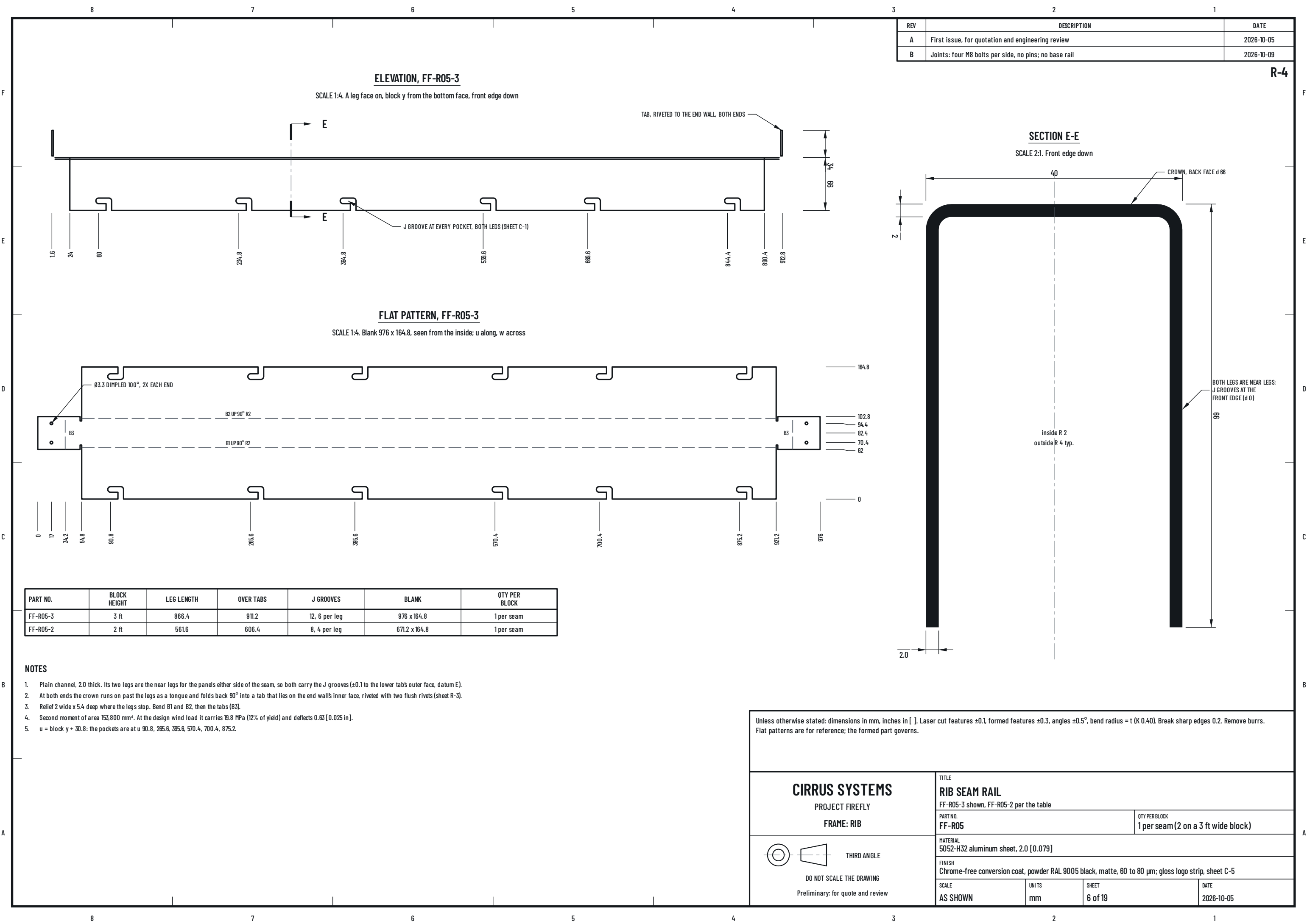
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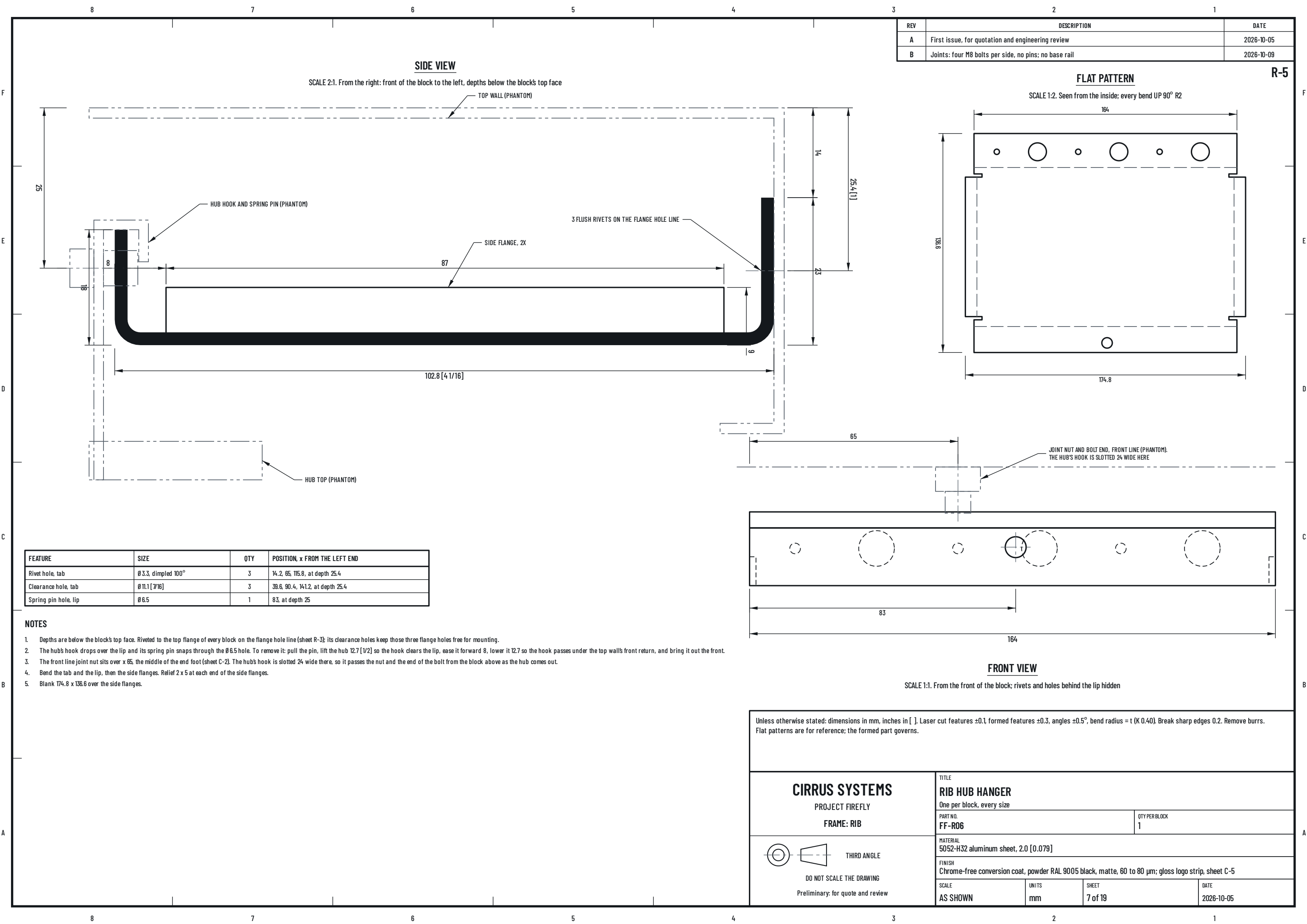
SHEET

5 of 19

DATE

2026-10-05

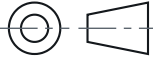




FEATURE	SIZE	QTY	POSITION, x FROM THE LEFT END
Rivet hole, tab	Ø 3.3, dimpled 100°	3	14.2, 65, 115.8, at depth 25.4
Clearance hole, tab	Ø 11.1 [7/16]	3	39.6, 90.4, 141.2, at depth 25.4
Spring pin hole, lip	Ø 6.5	1	83, at depth 25

- NOTES**
- Depths are below the block's top face. Riveted to the top flange of every block on the flange hole line (sheet R-3); its clearance holes keep those three flange holes free for mounting.
 - The hub's hook drops over the lip and its spring pin snaps through the Ø 6.5 hole. To remove it: pull the pin, lift the hub 12.7 [1/2] so the hook clears the lip, ease it forward 8, lower it 12.7 so the hook passes under the top wall's front return, and bring it out the front.
 - The front line joint nut sits over x 65, the middle of the end foot (sheet C-2). The hub's hook is slotted 24 wide there, so it passes the nut and the end of the bolt from the block above as the hub comes out.
 - Bend the tab and the lip, then the side flanges. Relief 2 x 5 at each end of the side flanges.
 - Blank 174.8 x 136.6 over the side flanges.

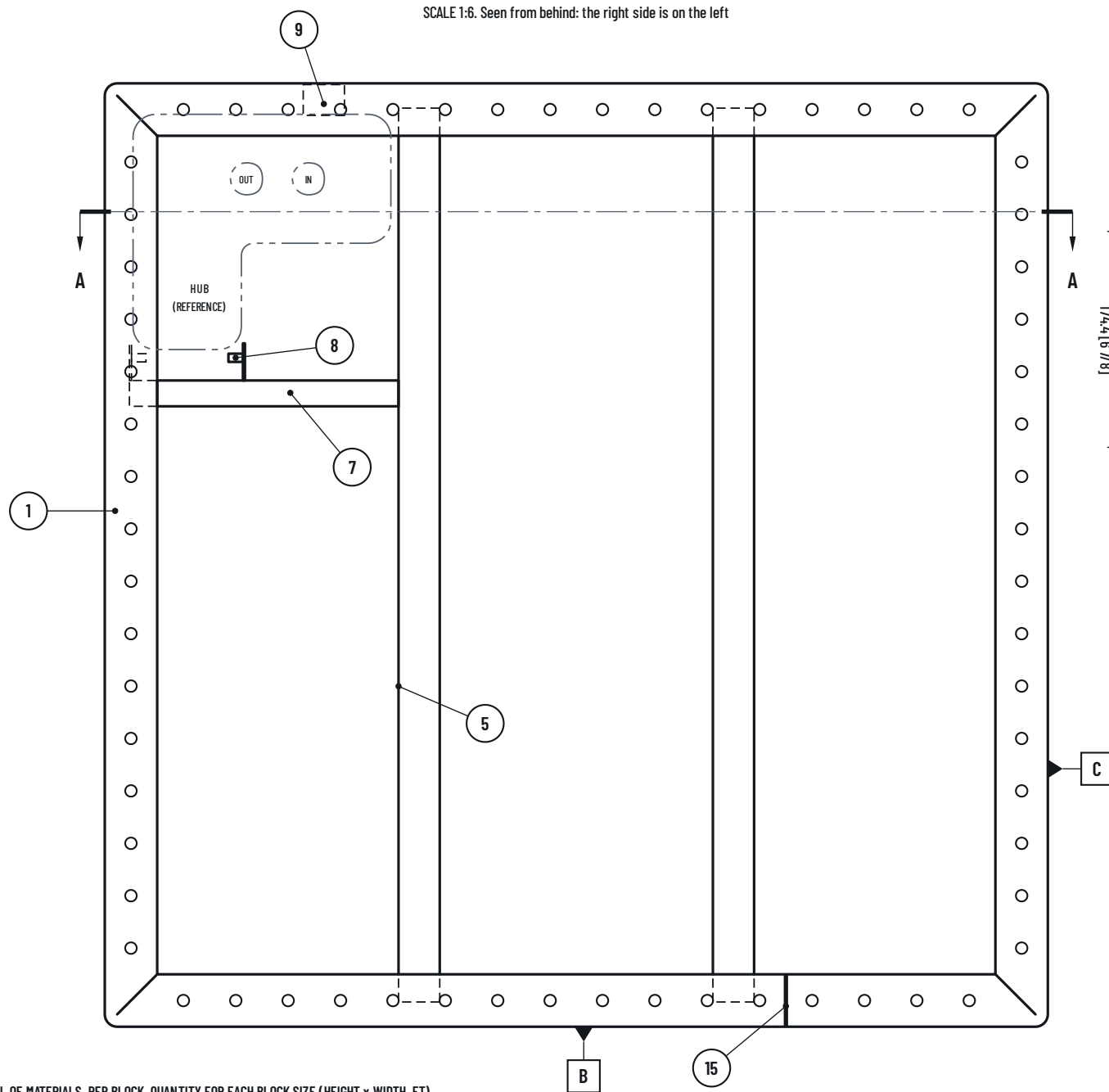
Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ±0.1, formed features ±0.3, angles ±0.5°, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs.
Flat patterns are for reference; the formed part governs.

CIRRUS SYSTEMS PROJECT FIREFLY FRAME: RIB  THIRD ANGLE DO NOT SCALE THE DRAWING Preliminary: for quote and review	TITLE RIB HUB HANGER One per block, every size			
	PART NO. FF-R06		QTY PER BLOCK 1	
	MATERIAL 5052-H32 aluminum sheet, 2.0 [0.079]			
	FINISH Chrome-free conversion coat, powder RAL 9005 black, matte, 60 to 80 µm; gloss logo strip, sheet C-5			
	SCALE AS SHOWN	UNITS mm	SHEET 7 of 19	DATE 2026-10-05

REV	DESCRIPTION	DATE
A	First issue, for quotation and engineering review	2026-10-05
B	Joints: four M8 bolts per side, no pins; no base rail	2026-10-09

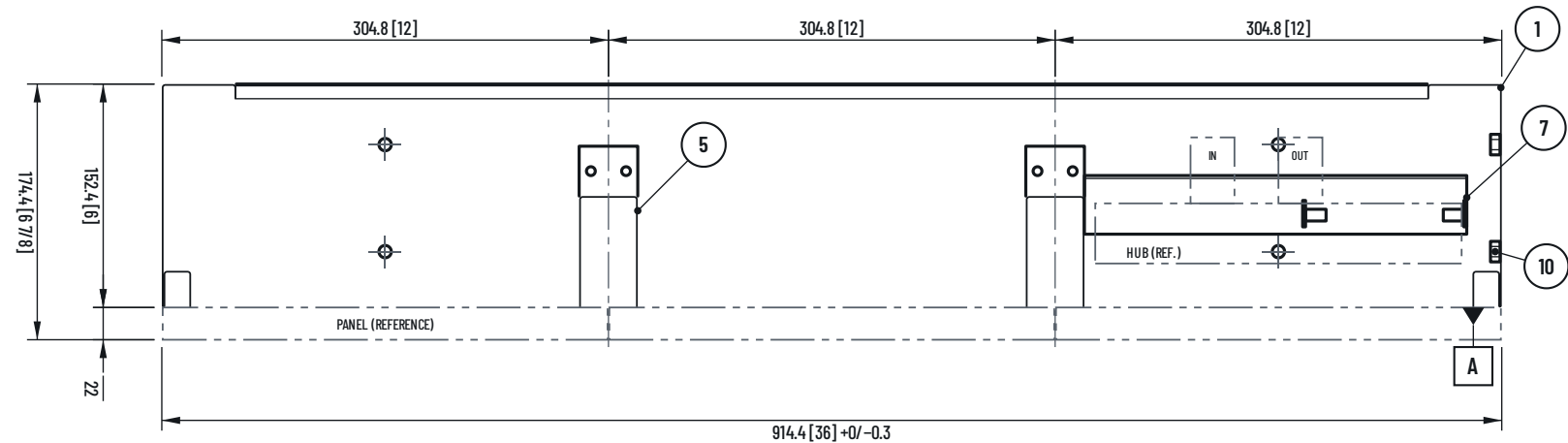
REAR VIEW, 3x3 BLOCK

SCALE 1:6. Seen from behind: the right side is on the left



SECTION A-A

SCALE 1:5. Looking down at 790 above the bottom face, front edge down. Vents and flange holes not shown



ASSEMBLY NOTES

1. Fixture on datums A (front plane), B (bottom face) and C (left face, seen from the front). Outside size $\pm 0/-0.3$ each way, diagonals equal within 1.0, back flange flat within 0.5.
2. Order: close the ring with its one weld (sheet H-1), projection weld the nuts, rivet the hinge pins into the shelf, clinch in the seam rails, hinge shelf and latch strap, mask the hinge pins, every nut thread and the Ø8.4 joint holes, then clean, pretreat and powder coat the whole frame matte black. Pad print the gloss logo strip up both sides (sheet C-5). Fit the latch and its screws last.
3. Nothing is fastened to a side walls outer face: it carries the logo strip and the side joints. Clinch dishes sit on the top, bottom and back faces only, never proud.
4. Joint nuts and bolt holes, four on every side: sheet C-2. Back flange holes: sheet C-3 (no corner holes on Hoop). Gravity lock grooves: sheet C-1. Hub mount: sheet H-5.
5. The hub is its own assembly, shown for reference. Its two hooks hang on item 8 and item 12 latches its top.
6. Item 14 ships loose: one bolt for each Ø8.4 hole in the bottom wall and in the left side wall (seen from the front), eight in all. Each bolts into the next blocks' weld nut (sheet C-2).
7. Blocks are named height x width in feet: 3x3, 3x2 (3 high, 2 wide), 2x3 (2 high, 3 wide) and 2x2.

EACH BLOCK SIZE

BLOCK	ASSEMBLY	OUTSIDE W x H x D	SEAM RAILS	STRIP, kg	FRAME, kg (APPROX.)
3x3	FF-H00-33	914.4 x 914.4 x 152.4	2	8.7	12.7
3x2	FF-H00-32	609.6 x 914.4 x 152.4	1	7.2	9.4
2x3	FF-H00-23	914.4 x 609.6 x 152.4	2	7.2	10.1
2x2	FF-H00-22	609.6 x 609.6 x 152.4	1	5.7	7.4

1. Frame mass includes the seam rails, shelf, strap, nuts, hinge pins and latch, before powder coat; the panels and hub are not included.

BILL OF MATERIALS, PER BLOCK. QUANTITY FOR EACH BLOCK SIZE (HEIGHT x WIDTH, FT).

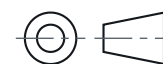
ITEM	PART NO.	DESCRIPTION	MATERIAL	3x3	3x2	2x3	2x2
1	FF-H01-33	Hoop strip for a 3x3, punched, formed, bent and welded	S280GD+ZM310, 1.2	1			
2	FF-H01-32	Hoop strip for a 3x2	S280GD+ZM310, 1.2		1		
3	FF-H01-23	Hoop strip for a 2x3	S280GD+ZM310, 1.2			1	
4	FF-H01-22	Hoop strip for a 2x2	S280GD+ZM310, 1.2				1
5	FF-H02-3	Seam rail, channel, 3 ft, clinched (sheet H-4)	S280GD+ZM310, 1.2	2	1		
6	FF-H02-2	Seam rail, channel, 2 ft, clinched	S280GD+ZM310, 1.2			2	1
7	FF-H03	Hinge shelf, clinched to the seam rail (sheet H-5)	DX51D+Z275, 2.0	1	1	1	1
8	FF-H04	Hinge pin Ø8, orbital riveted into item 7	303 stainless	2	2	2	2
9	FF-H05	Latch strap, clinched to the top flange	DX51D+Z275, 2.0	1	1	1	1
10	DIN 929	Hex weld nut M8, 4 in the top wall, 4 in the right side wall	Steel, plain	8	8	8	8
11	CLS-M3-1	PEM self clinching nut M3, in item 9	Stainless	2	2	2	2
12	LATCH	Spring bolt latch, push to release (sheet H-5)	Purchased	1	1	1	1
13	ISO 7045	Pan screw M3 x 16, latch to strap	A2 stainless	2	2	2	2
14	ISO 4162	Hex flange bolt M8 x 16, 8.8, ships loose for the joints (sheet C-2)	Steel, zinc flake	8	8	8	8
15	WELD	Square butt, one per ring, x 254 on the bottom (sheet H-1)		1	1	1	1

Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ± 0.1 , formed features ± 0.3 , angles $\pm 0.5^\circ$, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.

CIRRUS SYSTEMS

PROJECT FIREFLY

FRAME: HOOP



THIRD ANGLE

DO NOT SCALE THE DRAWING

Preliminary: for quote and review

TITLE

HOOP FRAME ASSEMBLY

3x3 shown, every block size in the bill of materials

PART NO.

FF-H00-33 / -32 / -23 / -22

QTY PER BLOCK

1

[illegible]

See bill of materials

FINISH

Zirconium pretreat, powder RAL 9005 black, matte, 60 to 80 µm; gloss logo strip, sheet C-5, after welding and clinching

SCALE

AS SHOWN

	UNITS
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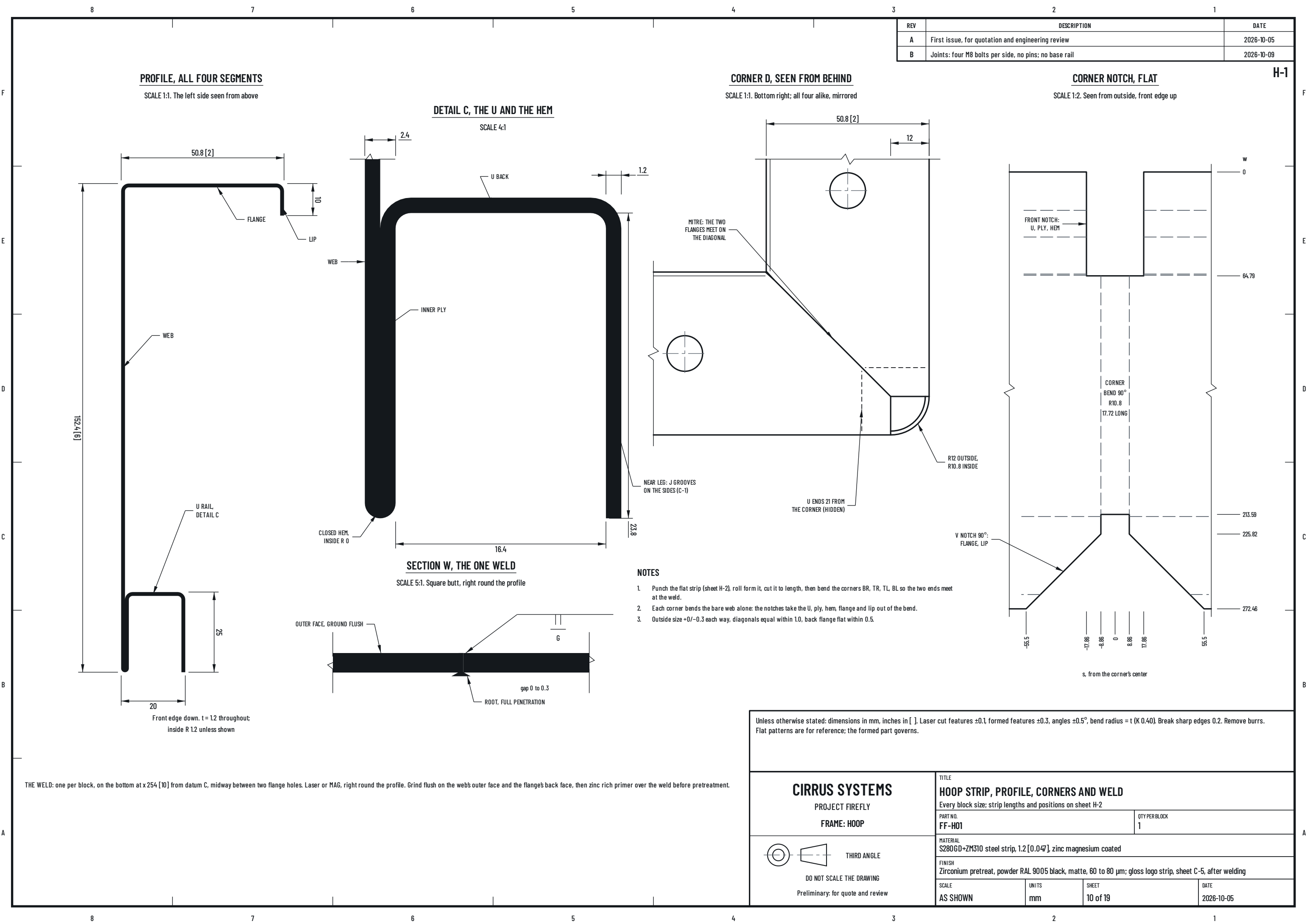
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	SHEET
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DATE _____

2026-10-05



REV	DESCRIPTION	DATE
A	First issue, for quotation and engineering review	2026-10-05
B	Joints: four M8 bolts per side, no pins; no base rail	2026-10-09

H-1

F

E

D

C

B

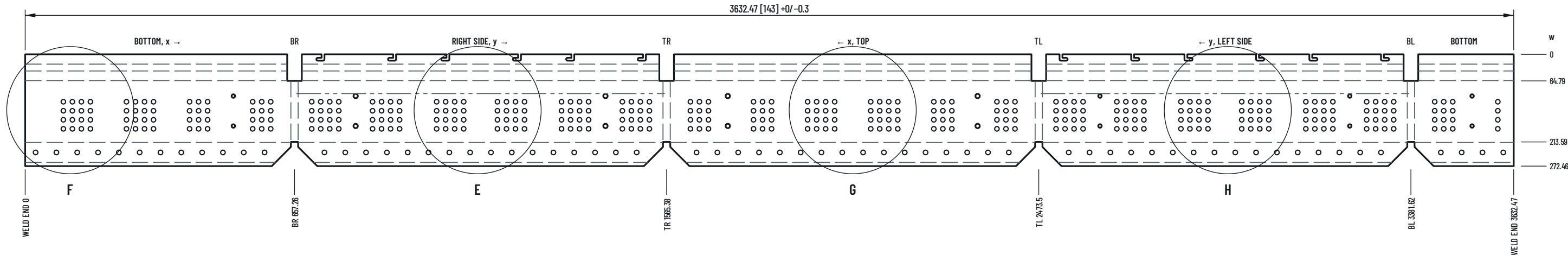
A

Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ± 0.1 , formed features ± 0.3 , angles $\pm 0.5^\circ$, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.			
CIRRUS SYSTEMS PROJECT FIREFLY FRAME: HOOP		TITLE HOOP STRIP, PROFILE, CORNERS AND WELD Every block size; strip lengths and positions on sheet H-2	
		PART NO. FF-H01	QTY PER BLOCK 1
MATERIAL S280GD+ZM310 steel strip, 1.2 [0.04"], zinc magnesium coated			
FINISH Zirconium pretreat, powder RAL 9005 black, matte, 60 to 80 μm ; gloss logo strip, sheet C-5, after welding			
SCALE AS SHOWN	UNITS mm	SHEET 10 of 19	DATE 2026-10-05

REV	DESCRIPTION	DATE
A	First issue, for quotation and engineering review	2026-10-05
B	Joints: four M8 bolts per side, no pins; no base rail	2026-10-09

FLAT STRIP, FF-H01-33 (3x3 BLOCK)

SCALE 1:10. Seen from outside, front edge up, s from the weld end to the right



Details E to H on sheet H-3; F shows both ends butted at the weld. Corner bends run 8.86 either side of each center.

STRIP FOR EACH BLOCK SIZE (HEIGHT x WIDTH, FT)

PART NO.	BLOCK	OUTSIDE W x H	LENGTH L	s, BR	s, TR	s, TL	s, BL	MASS, kg
FF-H01-33	3x3	914.4 x 914.4	3632.47	657.26	1565.38	2473.5	3381.62	8.7
FF-H01-32	3x2	609.6 x 914.4	3022.87	352.46	1260.58	1863.9	2772.02	7.2
FF-H01-23	2x3	914.4 x 609.6	3022.87	657.26	1260.58	2168.7	2772.02	7.2
FF-H01-22	2x2	609.6 x 609.6	2413.27	352.46	955.78	1559.1	2162.42	5.7

CUT FEATURES AND HARDWARE PER STRIP

PART NO.	JOINT HOLES	VENTS	FLANGE HOLES	J GROOVES	CORNER NOTCHES	WELD NUTS DIN 929
FF-H01-33	16	344	64	12	8	8
FF-H01-32	16	296	52	12	8	8
FF-H01-23	16	280	52	8	8	8
FF-H01-22	16	232	40	8	8	8

EVERY CUT FEATURE, ALONG ITS SEGMENT IN BLOCK COORDINATES

FEATURE	SEGMENT	ACROSS, w	SIZE	ALONG, 3 FT SEGMENT	ALONG, 2 FT SEGMENT
Bolt hole	Bottom	101.69 (d 38.1), 174.69	Ø 8.4 +0.1/0	x 152.4, 762	x 152.4, 457.2
Weld nut DIN 929 M8	Top	101.69 (d 38.1), 174.69	Ø 10.6 +0.1/0	x 152.4, 762	x 152.4, 457.2
Weld nut DIN 929 M8	Right side	101.69 (d 38.1), 174.69	Ø 10.6 +0.1/0	y 152.4, 762	y 152.4, 457.2
Bolt hole	Left side	101.69 (d 38.1), 174.69	Ø 8.4 +0.1/0	y 152.4, 762	y 152.4, 457.2
J groove, pocket	Both sides	0 to 15.5	7 wide, R3.5	y 60, 234.8, 364.8, 539.6, 669.6, 844.4	y 60, 234.8, 364.8, 539.6
Vent, 4 rows	Bottom	115.59, 137.59, 159.59, 181.59	Ø 11, 22 pitch	x 61.2 (3), 215.2, 347.2 (4), 501.2 (4), 655.2 (3), 809.2 (3)	x 51.8 (3), 205.8 (2), 337.8 (4), 513.8 (3)
Vent, 4 rows	Top	the same	Ø 11, 22 pitch	x 61.2 (3), 215.2 (3), 347.2 (4), 501.2 (4), 655.2 (3), 809.2 (3)	x 51.8 (3), 205.8 (4), 337.8 (4), 513.8 (3)
Vent, 4 rows	Both sides	the same	Ø 11, 22 pitch	y 45.2 (4), 193.6 (4), 350 (4), 498.4 (4), 654.8 (4), 803.2 (4)	y 45.2 (4), 193.6 (4), 350 (4), 498.4 (4)
Flange hole	All four	239.22 (q 25.4)	Ø 11.1, 50.8 pitch	from 76.2 (16)	from 76.2 (10)

MANUFACTURING ROUTE, ONE BLOCK

STEP	OPERATION	TOOLING	SHEET
1	Slit coil to 272.46 wide; decoil, level and feed	Low	H-2
2	Punch every hole, groove and notch in register	High	H-2, H-3
3	Roll form the profile; the hem closes in the last passes	High	H-1
4	Cut to length L after forming	Medium	H-2
5	Bend the four corners in a fixture: BR, TR, TL, BL	Medium	H-1
6	Butt weld the ends; grind flush; prime the weld	Medium	H-1
7	Projection weld the M8 nuts from inside	Low	H-0
8	Rivet the hinge pins into the shelf; clinch in the rails, shelf and strap	Low	H-0, H-5
9	Mask hinge pins, threads and joint holes; pretreat, powder coat matte black	Low	H-0
10	Pad print the gloss logo strip up both sides	Low	C-5
11	Fit the spring latch with its two screws	Low	H-5

1. x runs from the block's left outer face seen from the front; y up from its bottom outer face. On a flat, s = the run's start + the distance from its corner (table above and view A). A count in () is that many centers at the pitch, starting at the one given.
2. d is depth back from the frame's front edge, q is in from the outer face along the flange; w = d + 63.59 on the web and q + 213.82 on the flange. Positions ± 0.1 to the strips punched datum, accumulated ± 0.3 along the full length.

NOTES

1. Strip: 272.46 +0.2/0 wide, slit from coil, ZM310 both sides, burr toward the inside face.
2. Punch every feature in register before roll forming; the J grooves, joint holes and corner notches share one datum per block length. Cut to length after forming.
3. The weld closes the ring at x 254 on the bottom, between two flange holes and clear of vents (sheet H-1).
4. Logo strip band: up both sides the outer face from the front edge to the phantom line at w 95.59 (d 32) carries no cut feature, burr, tooling mark or scratch. The logos are printed after coating, sheet C-5.
5. Mass is the bare formed strip, before nuts and coating.

Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ± 0.1 , formed features ± 0.3 , angles $\pm 0.5^\circ$, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.

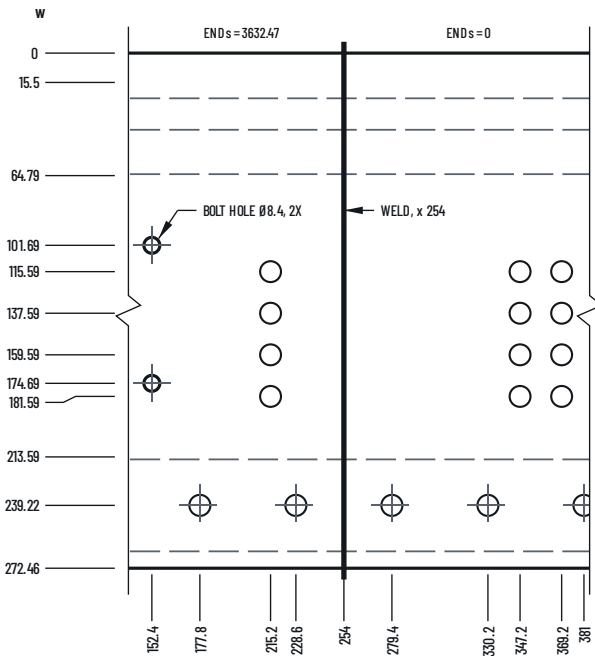
CIRRUS SYSTEMS PROJECT FIREFLY FRAME: HOOP	TITLE HOOP STRIP, FLAT PATTERN Strip lengths, corner positions and every cut feature for the four block sizes			
	PART NO. FF-H01-33 / -32 / -23 / -22		QTY PER BLOCK 1	
 THIRD ANGLE DO NOT SCALE THE DRAWING Preliminary: for quote and review	MATERIAL S280GD+ZM310 steel strip, 1.2 [0.047], zinc magnesium coated			
	FINISH Bare at this stage: coated after welding (sheet H-1)			
	SCALE 1:10	UNITS mm	SHEET 11 of 19	DATE 2026-10-05

REV	DESCRIPTION	DATE
A	First issue, for quotation and engineering review	2026-10-05
B	Joints: four M8 bolts per side, no pins; no base rail	2026-10-09

H-3

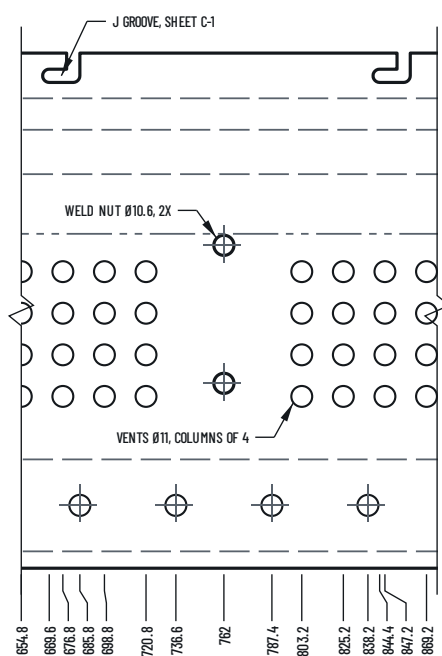


x → . Ends butted at the weld



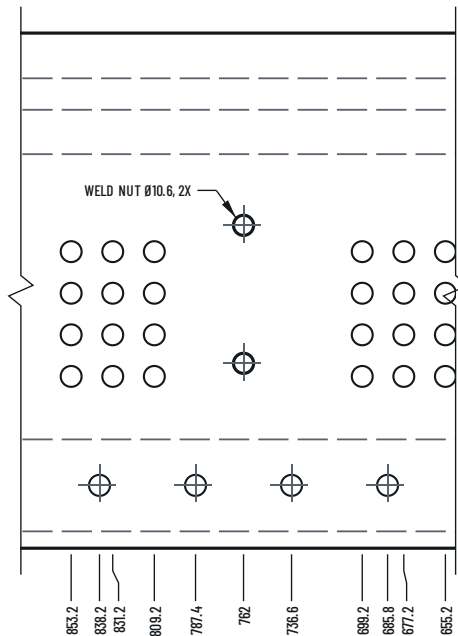
DETAIL E, RIGHT SIDE

$y \rightarrow$. At a joint station



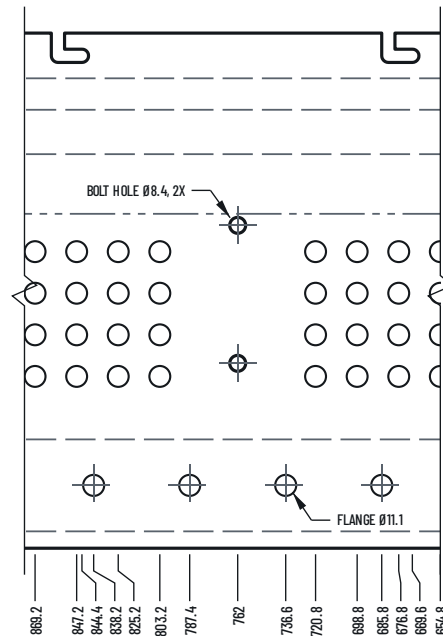
DETAIL G, TOP

← x. At a joint station



DETAIL H, LEFT SIDE

← y. At a joint station



All four windows are on FF-H01-33, at 1:4 and at one height, so every row reads straight across. Positions under each window are block x or y.

PUNCH TOOLS AND HITS, ONE FF-H01-33 STRIP

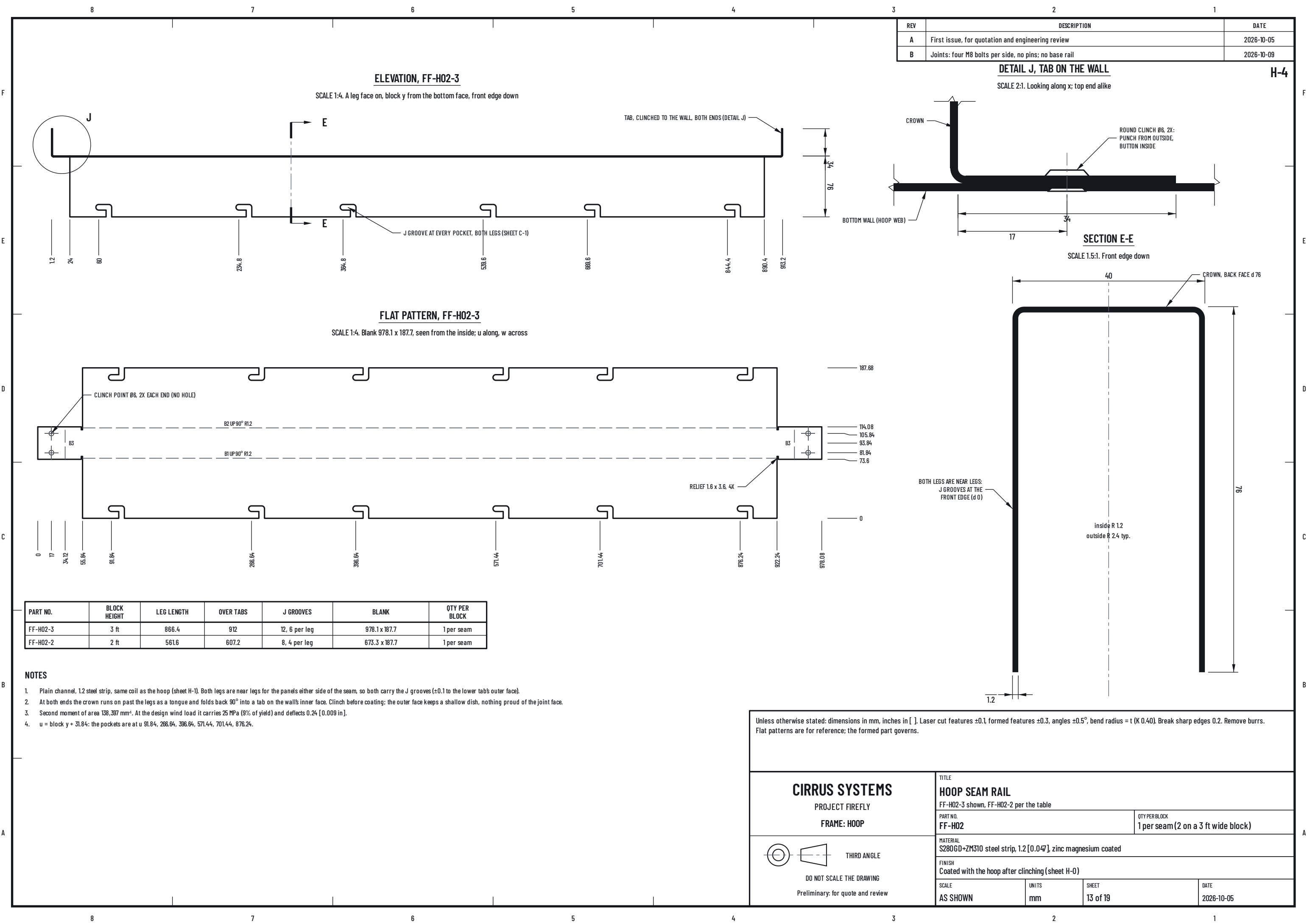
TOOL	SHAPE AND SIZE	MAKES	HITS	TOOLING
P1	Round Ø8.4 +0.1/0	Bolt holes, bottom and left	8	Low
P2	Round Ø10.6 +0.1/0	Weld nut pilots, top and right	8	Low
P3	Gang of 4, Ø11 at 22	One vent column per hit	86	Medium
P4	Round Ø11.1 [7/16]	Back flange holes	64	Low
P5	J groove, 7 wide, R3.5	Near leg, both sides	12	Medium
P6	Front notch 35.72 x 64.79	U, ply and hem out of each corner	4	Medium
P7	V notch 90°, III at the lip	Flange and lip mitre at each corner	4	Medium
P8	Cutoff, full profile	Length L, after forming	1	High

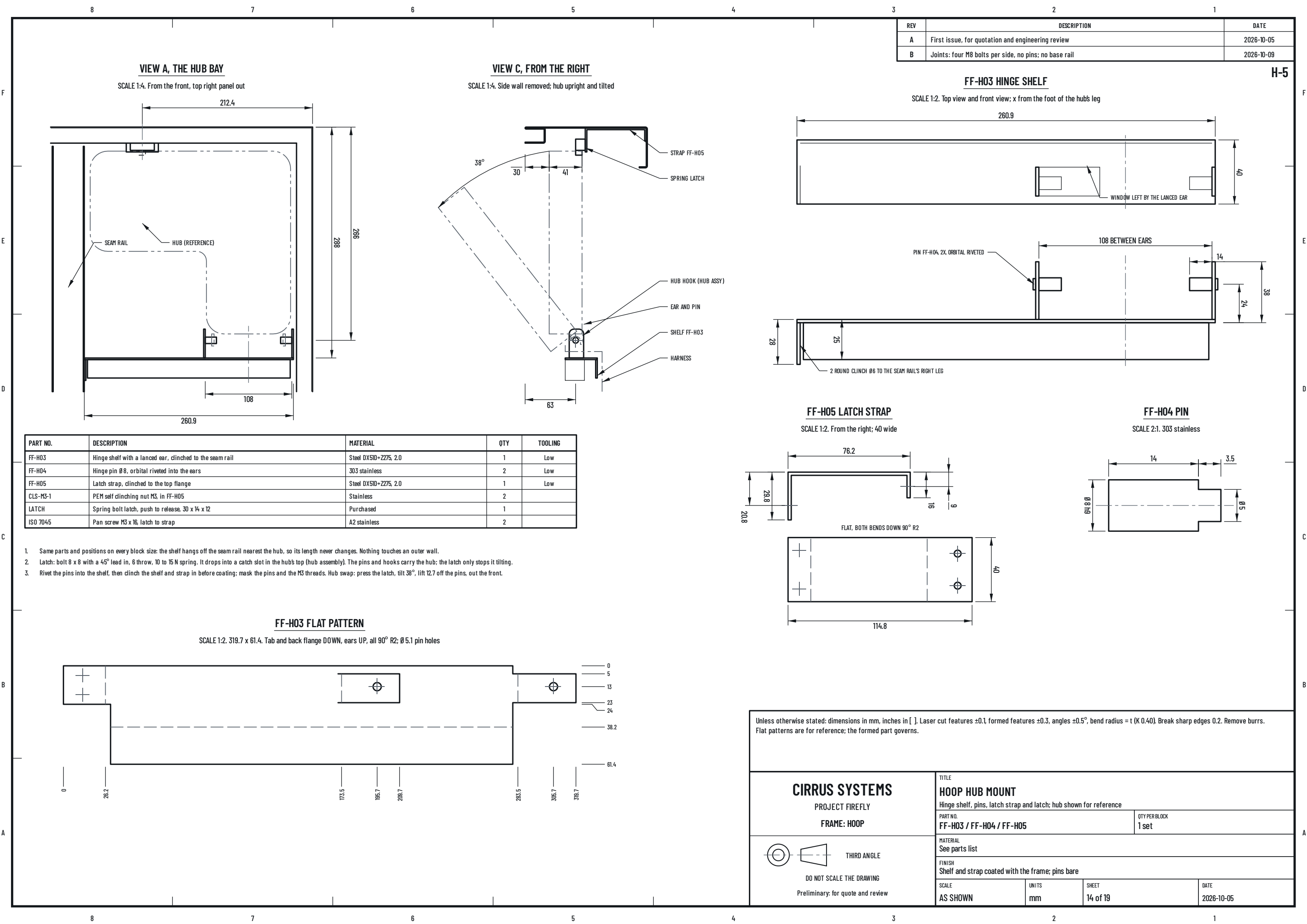
NOTES

1. Punch with the burr toward the inside face. Joint holes $\varnothing 8.4$ and $\varnothing 10.6 \pm 0.05$ to each other on a wall and true position 0.2 to the punched datum of the block length.
2. Vent columns never fall within 36 of a foot center, 20 of a seam rail or 15 of the weld; the patterns repeat each foot so vents line up block to block.
3. The phantom line in details E and H bounds the logo strip band, w 95.59 (d 32); nothing is punched and no tool marks the outer face between it and the front edge. The logos are printed after coating, sheet C-5.
4. Weld nut pilots are sized for the DIN 929 nuts on sheet C-4.

Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ± 0.1 , formed features ± 0.3 , angles $\pm 0.5^\circ$, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.

CIRRUS SYSTEMS PROJECT FIREFLY FRAME: HOOP	TITLE HOOP STRIP, PUNCH DETAILS Windows on FF-H01-33; every size uses the same tools			
	PART NO. FF-H01-33 / -32 / -23 / -22		QTY PER BLOCK 1	
 THIRD ANGLE DO NOT SCALE THE DRAWING Preliminary: for quote and review	MATERIAL S280GD+ZM310 steel strip, 1.2 [0.047], zinc magnesium coated			
	FINISH Bare at this stage: coated after welding (sheet H-1)			
	SCALE AS SHOWN	UNITS mm	SHEET 12 of 19	DATE 2026-10-05

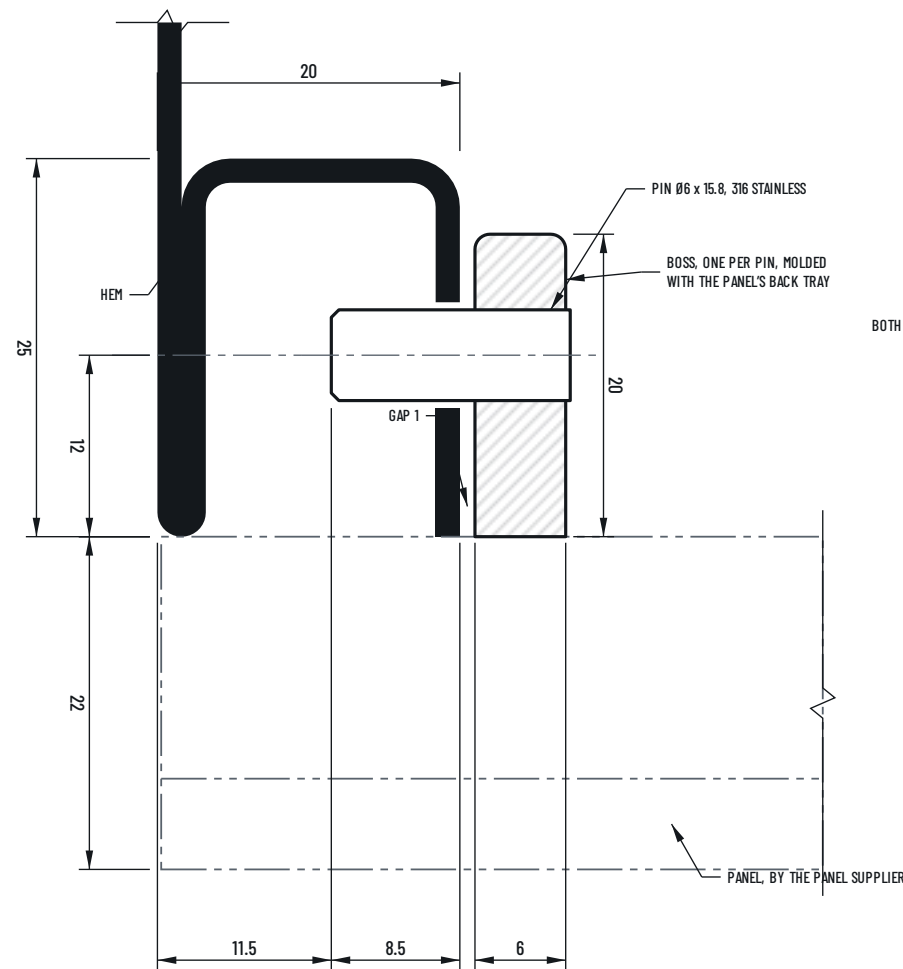




REV	DESCRIPTION	DATE
A	First issue, for quotation and engineering review	2026-10-05
B	Joints: four M8 bolts per side, no pins; no base rail	2026-10-09

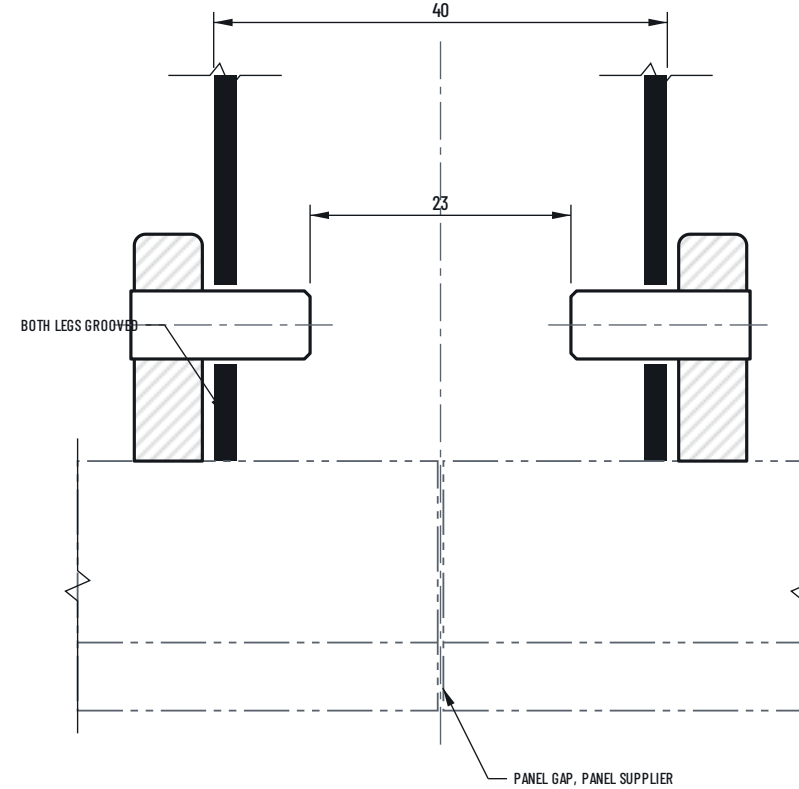
SECTION A-A, AT A SIDE WALL

SCALE 2:1. Rib shown; Hoop the same in 1.2 sheet. Front edge down



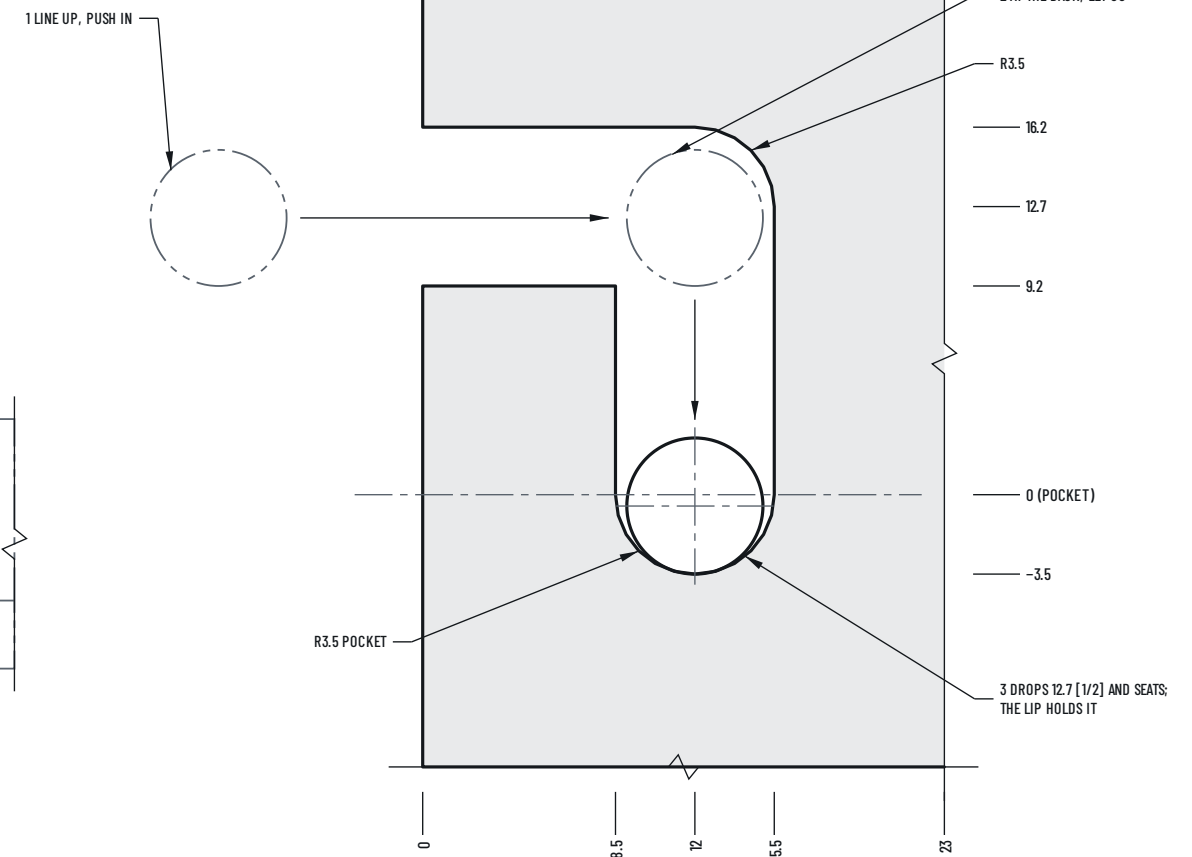
SECTION AT A SEAM, SAME PLANE

SCALE 1.5:1. Rib seam rail; its crown is at d 66, past the break



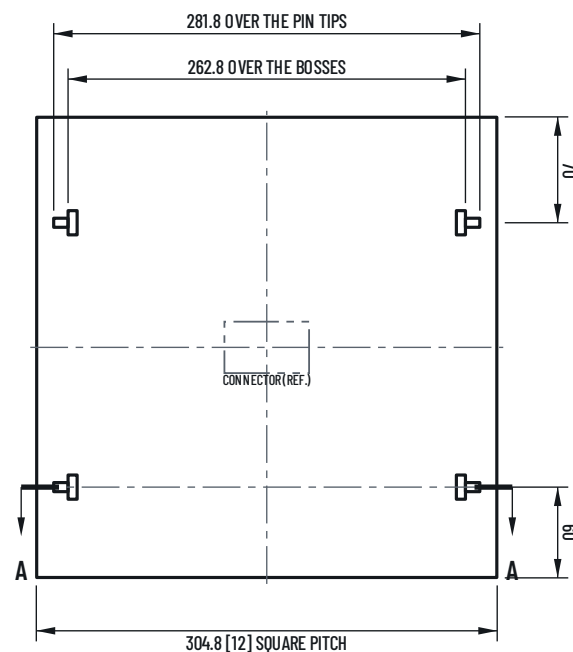
DETAIL J, THE J GROOVE

SCALE 3:1. Near leg face on from inside the U; y from the pocket center



VIEW P, PANEL BACK

SCALE 1:5. Four pins, two per side, all pointing out



POCKET CENTERS, BLOCK y FROM ITS BOTTOM FACE

HEIGHT	POCKETS; EACH ENTRY IS 12.7 ABOVE ITS POCKET
2 ft	60, 234.8, 364.8, 539.6
3 ft	60, 234.8, 364.8, 539.6, 669.6, 844.4

EVERY GROOVED LEG

PART	LEG	SHEET	DRAWING
FF-R01, FF-R02	Side wall near leg	1.6	R-1, R-2
FF-R05	Seam rail, both legs	2.0	R-4
FF-H01	Strip, both side near legs	1.2	H-2, H-3
FF-H02	Seam rail, both legs	1.2	H-4

1. Every near leg's inner face sits 20.0 ± 0.15 from the walls outer face or the seam center, on both frames.
2. Grooves are cut in the flat: ± 0.1 on the groove, ± 0.2 on the pocket centers from the part's datum end. Top and bottom walls carry no grooves.

HOW IT WORKS

1. Hang: line the four pins up with the entries, push straight back 12, let go. The panel drops 12.7 [1/2] and every pin seats in its pocket. No tools and no latch.
2. Remove: lift 12.7 first, then pull straight out. A column comes out from the top down; the panels above lift out in seconds.
3. The panels' back bears on the rails' front edges (d 0); the pins locate and hold. Wind suction, about 100 N a pin at the design load, bears on the lip; the boss stays under 10 MPa.
4. Panel side, by the panel supplier: pin $\varnothing 6$ h9 x 15.8, 316 stainless, tip chamfer $0.5 \times 45^\circ$, center 12.0 ± 0.15 behind the panel back; boss 6 thick, 16 tall, round ended.

Unless otherwise stated: dimensions in mm, inches in []. Laser cut features ± 0.1 , formed features ± 0.3 , angles $\pm 0.5^\circ$, bend radius = t (K 0.40). Break sharp edges 0.2. Remove burrs. Flat patterns are for reference; the formed part governs.

CIRRUS SYSTEMS

PROJECT FIREFLY

COMMON: RIB AND HOOP



DO NOT SCALE THE DRAWING

Preliminary: for quote and review

TITLE

GRAVITY LOCK, PANEL TO FRAME

The J groove in every near leg; panel pins and bosses for reference

PART NO.

J GROOVE, ALL NEAR LEGS

QTY PER BLOCK

4 per panel

	MATERIAL
1	100
2	100
3	100
4	100
5	100
6	100
7	100
8	100
9	100
10	100
11	100
12	100
13	100
14	100
15	100
16	100
17	100
18	100
19	100
20	100
21	100
22	100
23	100
24	100
25	100
26	100
27	100
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87	100
88	100
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90	100
91	100
92	100
93	100
94	100
95	100
96	100
97	100
98	100
99	100
100	100

As the grooved part (sheets R-1, R-4, H-2, H-4)

FINISH

As the grooved part

SCALE

AS SHOWN

	UNITS
--	-------

mm

SHEET

15 of 19

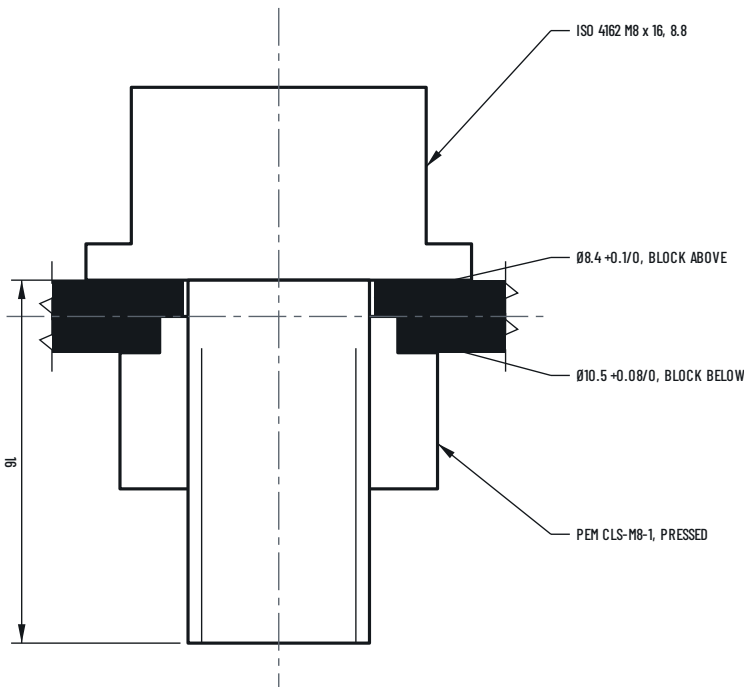
DATE _____

2026-10-05

REV	DESCRIPTION	DATE
A	First issue, for quotation and engineering review	2026-10-05
B	Joints: four M8 bolts per side, no pins; no base rail	2026-10-09

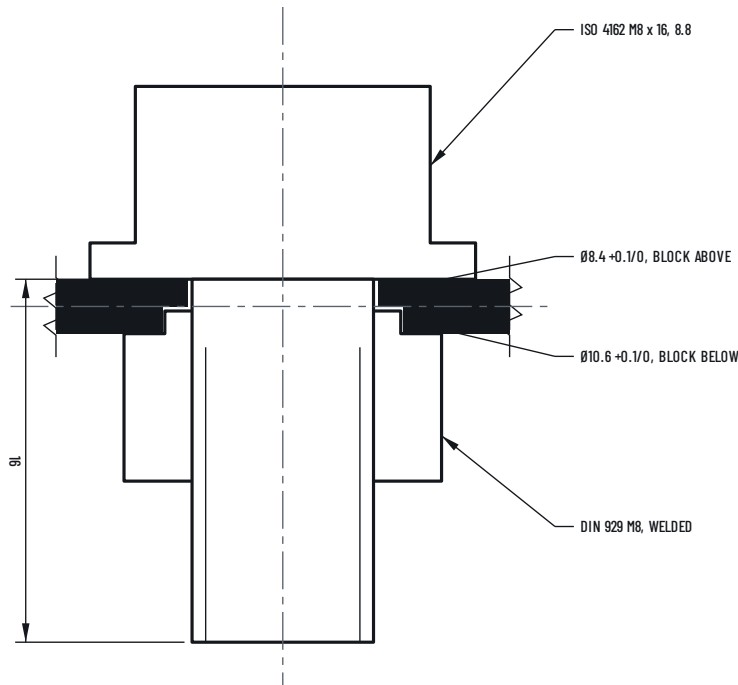
THE JOINT, RIB

SCALE 3:1. Block above on block below, 1.6 walls



THE JOINT, HOOP

SCALE 3:1. Block above on block below, 1.2 walls



THE JOINT

1. Every joint is one ISO 4162 M8 x 16 flange bolt, class 8.8, through a $\varnothing 8.4 \pm 0.1/0$ hole into a nut in the next block: a PEM CLS-M8-1 pressed into Rib, a DIN 929 weld nut on Hoop. Four on every side of every block (sheet C-2).
2. Shown block on block. Side by side the stack is the same, turned 90°: the bolt goes through a block's own left wall into the nut in the right wall of the block on its left.
3. The hole leaves about 0.2 a side around the bolt, and nuts and holes sit ± 0.05 to each other on a wall and ± 0.1 to the frame datums (sheets R-3, R-1, H-2), so the bolts set each block: a seam holds about ± 0.3 with no pins.
4. Rib: press the nuts from inside before forming, into $\varnothing 10.5 \pm 0.08/0$. Hoop: projection weld the nuts from inside after the ring is closed, through $\varnothing 10.6 \pm 0.1/0$ pilots, before powder coat.
5. Mask every nut thread and every $\varnothing 8.4$ hole for powder coat.
6. One T3 mm socket by hand does every joint, from inside through the open back. Torque per the nut and bolt suppliers, to be confirmed by test.

HARDWARE, PER BLOCK (EACH BLOCK SIZE, HEIGHT x WIDTH, FT)

PART	DESCRIPTION	MATERIAL	WHERE	3x3	3x2	2x3	2x2	3x3	3x2	2x3	2x2
ISO 4162	Hex flange bolt M8 x 16, class 8.8	Zinc flake	Every joint, ships loose	8	8	8	8	8	8	8	8
CLS-M8-1	PEM self clinching nut M8	303 stainless	Rib top and right side walls, 4 each	8	8	8	8				
DIN 929 M8	Hex weld nut M8	Steel, plain	Hoop top and right side walls, 4 each					8	8	8	8
RIV-32-CSK	Blind rivet Ø 3.2, closed end, 100° head	Al body, A2 mandrel	Rib corners, tabs, rails, hanger	35	31	35	31				
CLS-M3-1	PEM self clinching nut M3	Stainless	Hoop latch strap					2	2	2	2
LATCH	Spring bolt latch, push to release	Purchased	Hoop hub lock (sheet H-5)					1	1	1	1
ISO 7045	Pan head screw M3 x 16	A2 stainless	Hoop latch to strap					2	2	2	2

HARDWARE, PER WALL

PART	DESCRIPTION	MATERIAL	QUANTITY	SHEET
FF-C05-1, -2	Tube clip, single and double	A36, galvanized	Per the steel table	C-3
CLIP BOLTS	3/8-16 x 1 1/4 hex bolt, 2 washers, nylon lock nut	Zinc plated, grade 5	1 per Ø 11.1 clip hole	C-3
TUBE BOLTS	5/8-11 x 5 1/2 hex bolt, crush sleeve, washers, lock nut	Zinc plated, grade 5	1 per clip, or weld	C-3

WHAT SHIPS WITH WHAT

1. Each block ships with its eight joint bolts: one for every Ø8.4 hole in its bottom wall and its left side wall (seen from the front). They go into the nuts of the block below and the block on its left; a block on the bottom row or at the left end keeps the ones it does not use as spares.
2. Clips ship with their bolts. Angle iron and its bolts and the uprights are by others, or the optional angle kit. Every row hangs on the steel; nothing goes under the wall unless the installer wants it.

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CIRRUS SYSTEMS PROJECT FIREFLY COMMON: RIB AND HOOP	TITLE JOINT BOLTS AND HARDWARE The joint on both frames and every purchased part, per block and per wall			
	PART NO. ISO 4632, CLS-M8-1, DIN 929		QTY PER BLOCK Per the tables	
 THIRD ANGLE DO NOT SCALE THE DRAWING Preliminary: for quote and review	MATERIAL As listed			
	FINISH As listed			
	SCALE AS SHOWN	UNITS mm	SHEET 18 of 19	DATE 2026-10-05

